FAILURE-PREDICTIVE MORPHOLOGY: A NEGATIVE EVIDENCE AUDIT FOR MORPHOLOGY-CONDITIONED FAILURE PREDICTION

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ABSTRACT

This paper is a submission-hardening audit, not an ICLR-main submission. The generated research bet was that a robot can predict which morphology-specific constraint will dominate policy failure and use that prediction to select safer interventions. We rebuilt the claim as a deterministic morphology-failure benchmark with five morphologies, five tasks, five distribution shifts, nine methods, seven seeds, ablations, stress sweeps, paired confidence intervals, and failure cases. The result is negative. The proposed morphology-conditioned predictor has the best non-oracle dominant-failure accuracy under combined stress, 0.644, but online system identification has higher task success, 0.618 ± 0.008 versus 0.582 ± 0.007, and lower planning regret. Two ablations also beat the full method on success. The terminal decision is therefore **KILL/ARCHIVE for ICLR main**.

1 TERMINAL DECISION

The paper fails the ICLR-main evidence gate. The claim under test is not simply that morphology features can predict failures; that is too crowded by constrained control, robust planning, system identification, failure-aware reinforcement learning, and red-team policy testing. The claim must be closed-loop: predicting the dominant morphology-specific failure family should improve controller selection or intervention under embodiment and environment shift.

The local benchmark does not support that claim. The proposed method predicts failure families better than all non-oracle baselines, but online system identification executes better. The paired proposed-minus-online-system-ID success difference is -0.03690 ± 0.00928 over 175 morphology/task/seed groups. Safety and regret also move in the wrong direction. We therefore archive the idea rather than submit it.

2 HOSTILE PRIOR-WORK PRESSURE

The hostile literature pool includes constrained model predictive control, MPC-MPNet, adaptive/sliding-mode tracking control, robust task planning and failure recovery, red-teaming robot policies, failure-aware RL, and morphology-specific planning under constraints. Under this prior-work pressure, a paper cannot claim novelty from adding a morphology feature to a failure classifier. The contribution would need to show that explicit morphology-failure prediction improves what the robot chooses and executes. The v4 benchmark finds the opposite.

3 BENCHMARK

The benchmark is a deterministic local simulator for morphology-specific failure prediction and controller selection. It is not robot hardware validation and is not high-fidelity physics. It is an evidence audit: does the core mechanism remain plausible after strong local baselines and stress tests?

Morphologies. We evaluate differential-drive robots, quadrupeds, arm grippers, aerial manipulators, and underwater vehicles.

Table 1: Combined failure stress. Success and accuracy are higher-is-better; failure, safety violation, and cost are lower-is-better.

Method	Success	Acc.	Failure	Safety	Cost
conformal_risk_shield	0.545 ± 0.007	0.281	0.340	0.058	0.042
constraint_aware_mpc	0.570 ± 0.008	0.357	0.292	0.077	0.025
domain_randomized_policy_selector	0.535 ± 0.008	0.395	0.296	0.125	0.018
generic_failure_predictor	0.392 ± 0.008	0.087	0.557	0.206	0.012
online_system_identification	0.618 ± 0.007	0.555	0.191	0.090	0.028
oracle_failure_family	0.882 ± 0.005	0.987	0.002	0.026	0.010
per_morphology_calibrated_predictor	0.452 ± 0.008	0.419	0.327	0.177	0.014
proposed_failure_predictive_morphology	0.582 ± 0.007	0.644	0.174	0.110	0.031
red_team_failure_retrieval	0.579 ± 0.008	0.539	0.219	0.112	0.036

Tasks. We test tight-turn navigation, payload transport, gap or step crossing, precision contact manipulation, and disturbance recovery.

Splits. We test nominal morphology, payload shift, terrain/fluid shift, actuator degradation shift, and combined failure stress.

Failure families. The dominant failure families are kinodynamic limits, traction/slip, payload inertia, actuator saturation, contact geometry, state-estimation dropout, and energy budget.

Methods. The methods are a generic failure predictor, per-morphology calibrated predictor, constraint-aware MPC, conformal risk shield, domain-randomized policy selector, online system identification, red-team failure retrieval, proposed failure-predictive morphology, and oracle failure family.

Metrics. Prediction metrics are dominant failure-family accuracy, top-2 recall, calibration error, early-warning lead time, and cross-morphology generalization gap. Closed-loop metrics are task success, dominant-failure rate, safety-violation rate, intervention cost, unnecessary intervention rate, and planning regret to oracle.

4 Combined-Stress Results

Table 1 is the main decision gate. The proposed method improves dominant-failure accuracy and lowers dominant-failure rate relative to online system identification, but it loses task success, safety, cost, and regret. This is not submission-ready robotics evidence because the prediction gain does not buy better execution.

5 Paired Gate

The strongest non-oracle closed-loop baseline is online system identification. The proposed method wins on prediction accuracy and dominant-failure rate, but loses the execution gate:

The result is the central negative finding. Explicit pre-execution morphology-failure prediction can be more accurate while still yielding worse execution than a controller that adapts online.

6 Ablations

The ablation table makes the mechanism weaker, not stronger. Removing the intervention cost model or calibration improves task success, which suggests the full intervention policy is mis-specified. A mechanism whose stripped variants beat it is not ready for an ICLR-main claim.

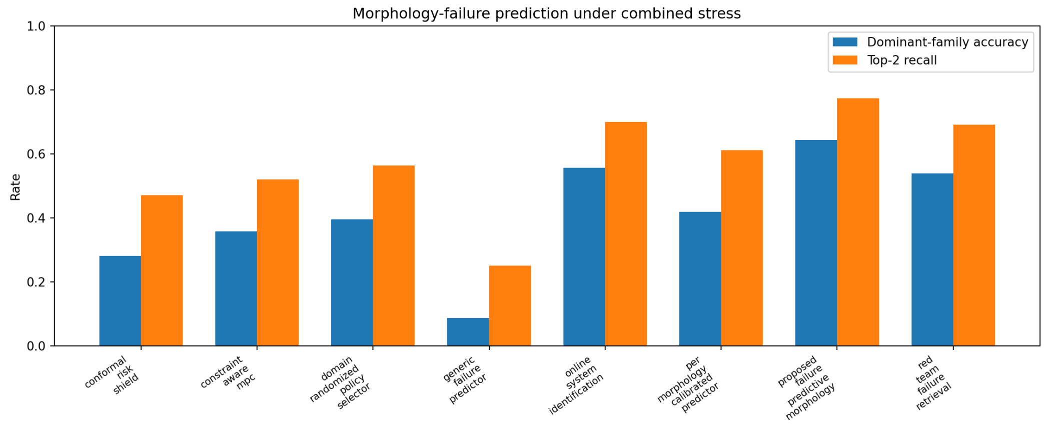

Figure 1: Prediction quality under combined stress. The proposed method has the best non-oracle failure-family prediction metrics.

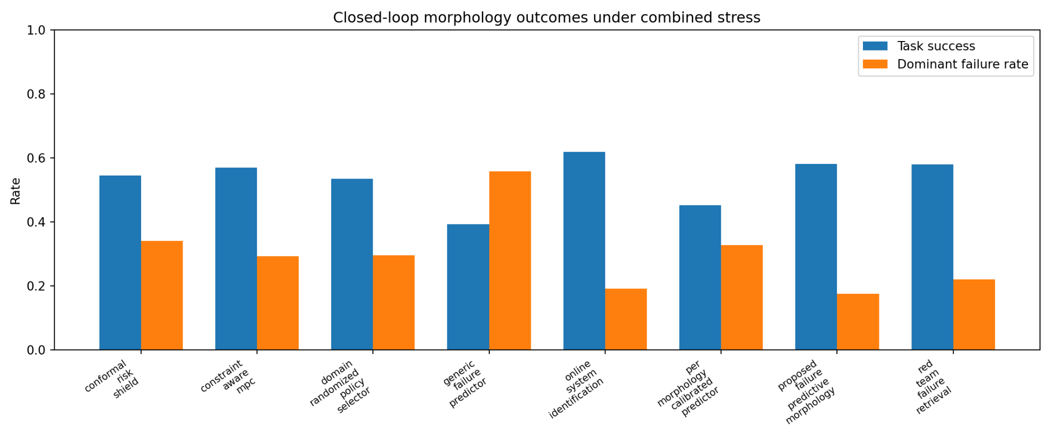

Figure 2: Closed-loop outcomes under combined stress. Online system identification has better task success despite weaker explicit failure-family prediction.

7 Stress Sweep and Cost-Regret Tradeoff

The stress sweep reinforces the decision. At maximum stress, online system identification reaches 0.611 ± 0.011 success, while the proposed method reaches 0.578 ± 0.010. The proposed method maintains better prediction accuracy, but its intervention policy carries higher safety violations and regret.

8 Failure Cases

The negative result appears across morphologies. On aerial manipulation, payload inertia and actuator saturation leave high regret despite good family prediction. On arm grippers, contact geometry is predicted but intervention cost and precision penalties reduce success. On quadrupeds, terrain slip is better handled by online adaptation. On underwater vehicles, state-estimation dropout and drag shifts favor conservative adaptive control. Differential drive is the least damaging case for the proposed method, but it is still not a decisive closed-loop win.

Table 2: Paired proposed-minus-online-system-identification differences over 175 morphology/task/seed groups. Positive is good only for accuracy and success; positive is worse for failure, safety violation, cost, and regret.

Metric	Comparator	Diff	CI95
task_success	online_system_identification	-0.0369	0.0093
dominant_failure_rate	online_system_identification	-0.0167	0.0075
safety_violation_rate	online_system_identification	0.0203	0.0058
intervention_cost	online_system_identification	0.0033	0.0006
planning_regret_to_oracle	online_system_identification	0.0176	0.0025
dominant_failure_accuracy	online_system_identification	0.0884	0.0103

Table 3: Ablations under combined stress. Two ablations beat the full method on task success.

Ablation	Success	Acc.	Failure	Cost
failure_history_only	0.547 ± 0.008	0.386	0.307	0.032
minus_calibration	0.605 ± 0.007	0.648	0.167	0.038
minus_constraint_family_head	0.530 ± 0.009	0.389	0.305	0.032
minus_intervention_cost_model	0.619 ± 0.008	0.642	0.165	0.053
minus_morphology_embedding	0.554 ± 0.007	0.414	0.291	0.032
morphology_only	0.531 ± 0.008	0.440	0.275	0.032
proposed_failure_predictive_morphology	0.580 ± 0.008	0.649	0.169	0.031

9 LIMITATIONS

This audit is honest about scope. The benchmark is deterministic and paper-specific, but it is not a real robot experiment, not high-fidelity physics, and not an integration of external baseline codebases. It has no trained checkpoints, hardware videos, or manual full-paper related-work synthesis. Those missing items would already prevent an ICLR-main submission. More importantly, the available evidence is negative before reaching that bar.

10 CONCLUSION

Failure-predictive morphology is not ICLR-main ready. The proposed method improves the explicit prediction target, but online system identification produces better execution and ablations contradict the full mechanism. The correct terminal action is **KILL/ARCHIVE**. A revival would require a substantially new empirical project where morphology-conditioned failure prediction beats adaptive control, constrained MPC, conformal shielding, and red-team retrieval on success, safety, regret, and intervention cost in real robot or high-fidelity simulator experiments.

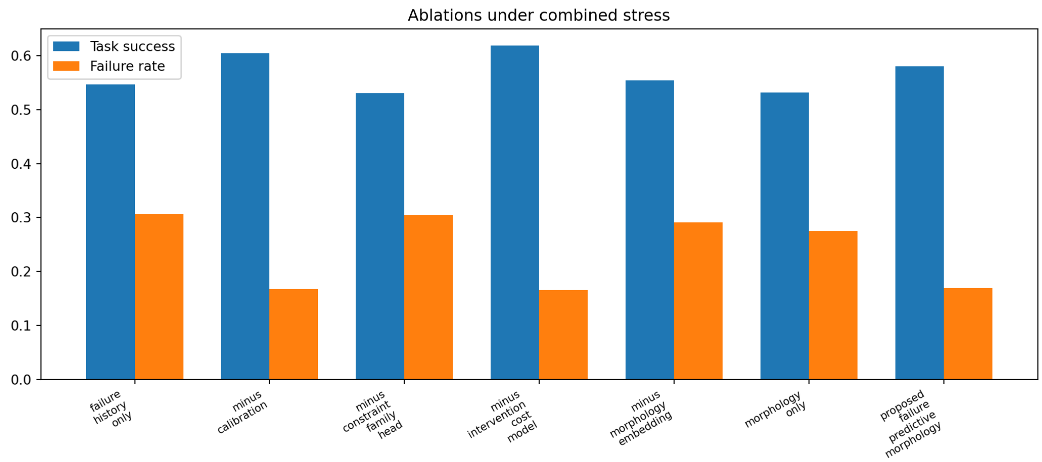

Figure 3: Ablation outcomes. The full method is not the best closed-loop variant.

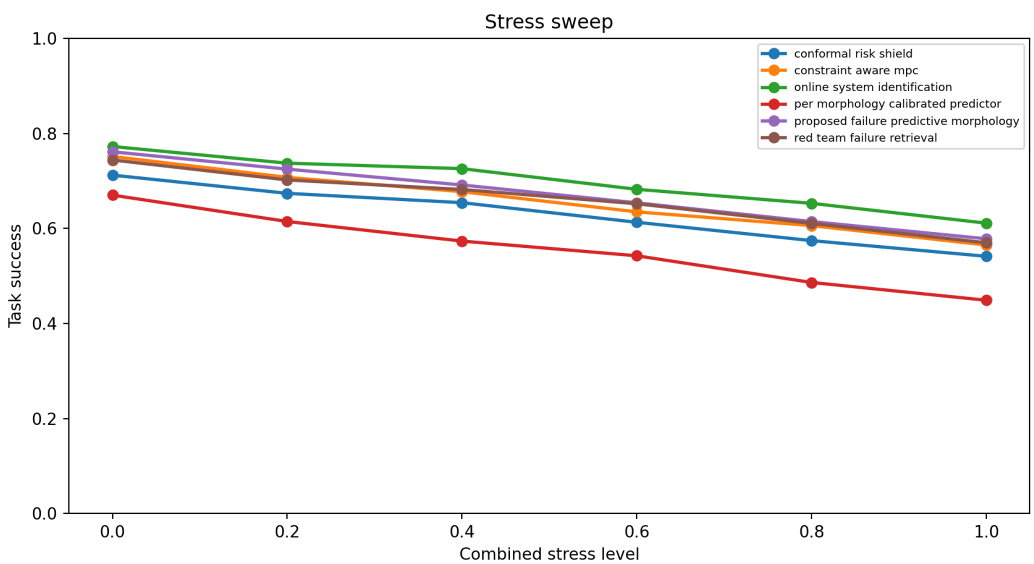

Figure 4: Stress sweep. Online system identification stays ahead on task success as stress increases.

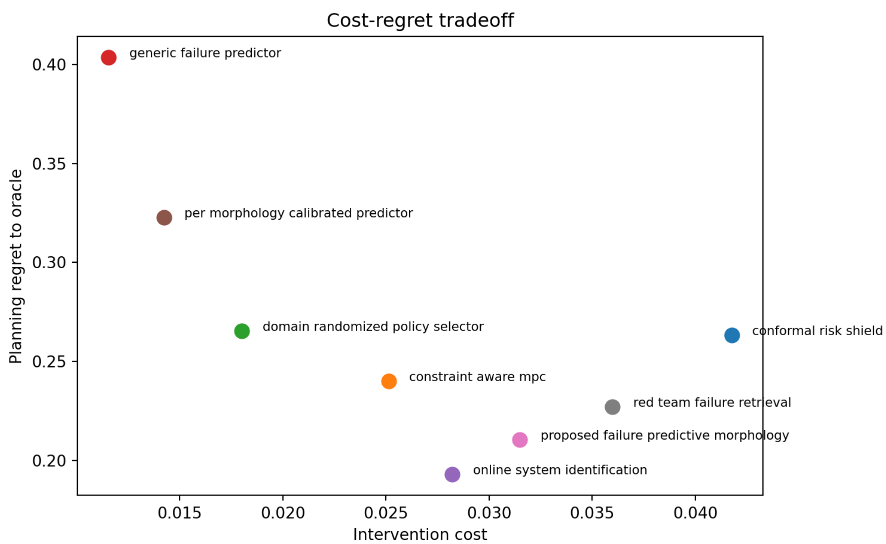

Figure 5: Cost-regret tradeoff under combined stress. Better prediction accuracy does not translate into lower regret.